

Hemant Parakh

Data Scientist · Forecasting & Applied Machine Learning

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PROFILE

Data Science M.Sc. candidate with two years at Hapag-Lloyd AG building production freight-rate forecasting models and LLM-powered analytics, across time-series forecasting, commercial data and applied GenAI.

PROFESSIONAL EXPERIENCE

Hapag-Lloyd AG - Working Student, Trade Analytics & Yield Management

Sep 2024 – Present

Hamburg, Germany · Trade Management Analytics (2024–2026), reorganised into Yield Management (2026–present)

- Freight-rate forecasting at 7/14/30-day horizons across global trade lanes, fusing internal segmentation data (TEU, NFR, CL2) with macro and market signals; panel models validated via walk-forward backtesting.
- Shipped an LLM trade-strategy pipeline turning a manager's idea into a structured volume-opportunity briefing (split research/writer prompts, prompt caching, match-confidence tiering), cutting analysis from 2-3 days to under an hour.
- Built a modular Python pipeline for external macro indicators (FRED, GDELT, container freight indices) with automated ingestion, cleaning and scheduled refresh.
- Migrated Databricks App LLM inference to Model Serving endpoints, removing a cold-start bottleneck (~90% faster load); integrated outputs into Qlik dashboards.

Gourav Mardia & Co - Intern, Big Data & Finance

Mar 2023 – Oct 2023

Mumbai, India

- Built relational databases and data-collection systems for financial reporting; applied statistical analysis to client data and turned findings into management recommendations.

APSIT Skills - Intern, Machine Learning

Jun 2021 – Aug 2021

Thane, India

- Built a real-time motion-detection system (background subtraction) tuned to reduce false positives across lighting conditions; published in an engineering research journal.

EDUCATION

Technische Universität Hamburg (TUHH) - M.Sc. Data Science

Oct 2023 – Dec 2026

Focus: AI · Big Data Analytics · Data Visualization · Causal Data Science.

A.P. Shah Institute of Technology - B.E. Computer Engineering

2019 – 2023

Thane, India · CGPA **8.86 / 10.0** (German equivalent: **1.5**)

MASTER'S THESIS

TU Hamburg × Hapag-Lloyd AG

Jun – Dec 2026

"Data-Driven Decision Support for Forecasting and Driver Analysis in Maritime Trade"

- Three questions: which macroeconomic drivers move maritime demand and freight rates, how large their effects are, and how stable those drivers stay across market regimes.
- Panel modelling across trade lanes on 5+ years of weekly data spanning multiple regimes (pre-COVID, COVID, peak, trough, Red Sea); benchmarks SARIMA/SARIMAX and Prophet against gradient boosting under walk-forward validation.
- SHAP-based driver attribution with stability analysis across windows, producing interpretable output for commercial decision-makers rather than a black-box forecast.

INDEPENDENT RESEARCH PROJECT

Systematic Literature Review - TU Hamburg

Jun – Oct 2025

"Consumer Perception of AI-Based Software: A Systematic Literature Review on Trust and Interface Design"

- PRISMA 2020 single-author review: screened 124 records to 17 included studies; built a two-cluster trust/interface-design framework and a five-dimension rubric rating each study's evidence strength.

SELECTED PROJECTS

- **"Diego"** - AI workplace conversation simulator (AI Beavers Hackathon 2026): real-time ElevenLabs voice pipeline with emotional-state modulation for rehearsing difficult conversations.
- **Cityblock** - animated web app (2025): interactive urban cross-section driven by live weather API data; front-end built end-to-end solo.
- **Student attentiveness** - detection in classrooms (2022): real-time model combining facial-expression and body-language analysis, integrated into an education platform.

TECHNICAL SKILLS

Languages & Tools: Python (pandas, NumPy, scikit-learn, statsmodels), SQL, Git, Bash, Selenium

ML & Statistics: Forecasting (SARIMA, Prophet), panel regression, gradient boosting, walk-forward validation, SHAP

GenAI & LLMs: Prompt engineering & caching, multi-agent prompts, RAG, MCP servers, LLM API integration

Platforms & BI: Databricks (Apps, Model Serving), AWS (Lambda), Power BI, Tableau, Qlik, matplotlib/Plotly

Research & Certs: PRISMA systematic reviews, evidence synthesis, LaTeX/APA; AWS, ML, SQL & BI certifications

Languages: English (C1), Hindi (C2), German (B1)